Type MS · Finding the equation of a normal

Answer 1 of 14 · Question 9 part whole · 5 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

9	$\frac{dy}{dx} = -(x^3 - 2x)e^{1-x} + (3x^2 - 2)e^{1-x}$	M1 A1
	When $x = 1$ $\frac{dy}{dx} = 2$ $\Rightarrow$ Gradient of **normal** $= -\frac{1}{"2"}$	M1
	$(y+1) = -\frac{1}{2}(x-1)$ oe and isw once seen	M1 A1 (5)
		Total 5 marks

Type MS · Finding the equation of a normal

Answer 1 of 14 · Question 9 part whole · 5 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 2 of 14 · Question 4 part whole · 10 marks

2019 JUN · Paper 01 (regional R) · find part whole in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding the equation of a normal

Answer 3 of 14 · Question 8 part c · 3 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

8(a)	$2xy+5y=e^x \quad y=\dfrac{e^x}{(2x+5)}$	
	$\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{e^x}{(2x+5)}\times\dfrac{(2x+5-2)}{(2x+5)}=\dfrac{y(2x+3)}{(2x+5)}$ *	M1A1 (5)
(b)	$x=0\Rightarrow \dfrac{dy}{dx}=\dfrac{5-2}{5^2}=\dfrac{3}{25}$	M1A1 (2)
ALT	$x=0\Rightarrow y=\dfrac{1}{5},\ \dfrac{dy}{dx}=\dfrac{1}{5}\times\dfrac{3}{5}=\dfrac{3}{25}$	
(c)	$x=0\Rightarrow y=\dfrac{e^0}{(2\times0+5)}=\dfrac{1}{5}$	M1(Award if seen in (b) and **used** in (c))
	$y-\dfrac{1}{5}=-\dfrac{25}{3}x$	M1
	$125x+15y-3=0$	A1 (3) [10]

Part	Mark	Notes
(a)		$2xy+5y=e^x \Rightarrow y=\dfrac{e^x}{(2x+5)}$
	M1	For attempting Quotient Rule • Both terms must be differentiated correctly $e^x\Rightarrow e^x \quad 2x+5\Rightarrow 2$ • There must be two terms subtracted in the numerator either way around • The denominator must the denominator squared. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$
	A1	For $e^x(2x+5)$ or $2e^x$
	A1	For a fully correct differentiated expression. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$

Type MS · Finding the equation of a normal

Answer 3 of 14 · Question 8 part c · 3 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

	M1	Subs in $y=\frac{e^x}{(2x+5)}$ as a common factor $\frac{dy}{dx}=\frac{e^x}{(2x+5)}\times\frac{(2x+5-2)}{(2x+5)}=\frac{y(2x+5-2)}{(2x+5)}$	Subs in $e^x=y(2x+5)$ and factorises $\frac{dy}{dx}=\frac{y(2x+5)(2x+5)-2y(2x+5)}{(2x+5)^2}$
	A1	For the correct answer with no errors. **Note this is a given answer.** $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
	ALT – uses implicit differentiation on $2xy+5y=e^x$		
	M1	$2\left(y+x\frac{dy}{dx}\right)+5\frac{dy}{dx}=e^x$	
	A1	Takes out $\frac{dy}{dx}$ as a common factor $\frac{dy}{dx}(2x+5)=e^x-2y$	
	A1	For a fully correct differentiated expression as below. $\frac{dy}{dx}=\frac{e^x-2y}{(2x+5)}$	
	M1	For separating the fraction, taking out y as a common factor and attempting to form a single fraction $\frac{dy}{dx}=\frac{e^x}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5)}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5-2)}{2x+5}$	
	A1	For the correct answer with no errors. $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
(b)	M1	For substituting $x=0$ into $\frac{dy}{dx}=\frac{e^x(2x+5)-2e^x}{(2x+5)^2}=\frac{e^0(2\times0+5)-2e^0}{(2\times0+5)^2}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
	ALT		
	M1	When $x=0\Rightarrow y=\frac{1}{5}$, $\frac{dy}{dx}=\frac{1}{5}\times\frac{3}{5}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
(c)	M1	$x=0\Rightarrow y=\frac{e^0}{(2\times0+5)}=\frac{1}{5}$ $x=0\Rightarrow y=\frac{1}{5}$ Award if seen in (b) and **used** in (c) **This is a B mark in Epen.**	
	M1	Inverts the gradient found in (b) and forms equation of the normal. ft their value of y $y-\frac{1}{5}=-\frac{25}{3}x$	
	A1	Equation of line is given in the required form. $125x+15y-3=0$	

Type MS · Finding the equation of a normal

Answer 3 of 14 · Question 8 part c · 3 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	$6^2 + x^2 - 12^2 \quad x^2 - 108$	

Type MS · Finding the equation of a normal

Answer 4 of 14 · Question 5 part b · 5 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
5(a)	$y = Q(x-6)(x+2)$ where Q is a constant Using the coordinates $(4,-6)$	M1
	$-6 = Q(4-6)(4+2) \Rightarrow Q = \frac{-6}{-12} = \frac{1}{2}$	M1
	$y = \frac{1}{2}(x-6)(x+2) \Rightarrow y = \frac{x^2}{2} - 2x - 6$ *	A1 cso [3]
(b)	$\frac{dy}{dx} = \frac{2x}{2} - 2$	M1
	$x = 4, \quad \frac{dy}{dx} = 4 - 2 = 2$	M1
	Gradient of normal is $-\frac{1}{2}$	B1ft
	Equation of l: $y - (-6) = -\frac{1}{2}(x-4)$	M1
	$\Rightarrow y = -\frac{x}{2} - 4 \Rightarrow 2y + x + 8 = 0$ *	A1 cso [5]
(c)	$\frac{x^2}{2} - 2x - 6 = -\frac{x}{2} - 4 \Rightarrow \frac{x^2}{2} - \frac{3}{2}x - 2 = 0$	M1
	$\Rightarrow (x-4)(x+1) = 0 \Rightarrow x = -1, 4$	M1 A1
	Area $= \int_{-1}^{4}\left(\frac{x^2}{2} - 2x - 6\right) dx - \int_{-1}^{4}\left(-\frac{x}{2} - 4\right) dx$	M1
	$= \left[\frac{x^3}{6} - \frac{3}{4}x^2 - 2x\right]_{-1}^{4}$	M1
	$= \left(\frac{4^3}{6} - \frac{3}{4}\times 4^2 - 2\times 4\right) - \left(\frac{(-1)^3}{6} - \frac{3}{4}\times(-1)^2 - 2\times(-1)\right)$	M1
	$= \left(\frac{4^3}{6} - \frac{3}{4}\times 4^2 - 2\times 4\right) - \left(\frac{(-1)^3}{6} - \frac{3}{4}\times(-1)^2 - 2\times(-1)\right)$	
	$= -\frac{125}{12} \Rightarrow$ Area $= \frac{125}{12}$ (units^2) oe	A1 [7]
		Total 15 marks

Part	Mark	Notes
(a)	M1	Uses the intersections with the x-axes to form a quadratic equation of the form $y = Q(x \pm 6)(x \pm 2)$

Type MS · Finding the equation of a normal

Answer 4 of 14 · Question 5 part b · 5 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

	M1	Uses their Quadratic with the coordinates $(4, -6)$ to find the value of Q Allow just one processing error here.
	A1 cso	For the correct equation in the required form. Note this equation is given to candidates. Both above steps must be complete and correct for the award of this mark.
	ALT – Uses simultaneous equations	
	M1	Sets up all three equations with the given coordinates. These must be correct. $y = px^2 + qx + r$ $0 = 4p - 2q + r$ **1** $0 = 36p + 6q + r$ **2** $-6 = 16p + 4q + r$ **3**
	M1	Attempts to solve their three simultaneous equations to find the values of p, q and r At least one correct value is evidence of correct method. **2 - 1** $0 = 32p + 8q$ **4** **3 - 2** $6 = 20p + 2q$ **5** **5** $\times 4$ $24 = 80p + 8q$ **6** **6 - 4** $24 = 48p \Rightarrow p = \frac{1}{2}$, $24 = 40 + 8q \Rightarrow q = -2$ $0 = 2 + 4 + r \Rightarrow r = -6$
	A1 cso	For the correct equation in the required form. Note this equation is given to candidates. All of the above steps must be complete and correct for the award of this mark.
(b)	M1	For differentiating the given expression. This must be correct, simplified or unsimplified for this mark.
	M1	For substituting $x = 4$ into their $\frac{dy}{dx}$ to find the gradient of the tangent.
	B1ft	For finding the gradient of the normal. Ft their gradient.
	M1	For forming an equation for l using the equation of the normal which must have come from use of calculus. If they use $y = mx + c$ then they must find c and form an equation for the award of this mark. For example; $c = -4 \Rightarrow y = -\frac{x}{2} - 4$
	A1 cso	For the correct equation in the required form. Accept the terms in any order. For example: even $0 = -8 - x - 2y$
(c)	M1	For equating the equation of S to their l and forming a 3TQ
	M1	For attempting to solve the 3TQ [see General Guidance for the definition of an attempt] to find two points of intersection.
	A1	For **both** correct values of x
	M1	For a correct statement for the area using **their two points of intersection** correctly. **Do not accept limits of $x = -2$ and 4 or 6** They may complete these two areas separately and combine at the end. Check to the end of their work before you score this mark. Accept either $\int_{-1}^{4} \text{Curve} - \int_{-1}^{4} \text{Line}$ or $\int_{-1}^{4} \text{Line} - \int_{-1}^{4} \text{Curve}$ for this mark.
	M1	For an attempt to integrate either the expression for the line or the curve.

Type MS · Finding the equation of a normal

Answer 4 of 14 · Question 5 part b · 5 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

		See General guidance – but no power of x is to decrease.
	M1	For substituting in their values correctly **AND** subtract the two integrals. Explicit substitution must be seen if the final area is incorrect, or the limits are incorrect. A final correct area which follows correct integration is adequate evidence.
	A1	For the correct area

Question	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 5 of 14 · Question 6 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 6 · find part c in this mark scheme extract

6 (a)	$x^2-5x+4=(x-4)(x-1)$	M1
	$A(1,0)$ $B(4,0)$	A1 A1 (3)
	Notes	
(a)		
M1	For solving the quadratic	
A1	For $A(1,0)$	
A1	For $B(4,0)$	
(b)	$\frac{dy}{dx}=2x-5$	M1
	When $x=1$ $\frac{dy}{dx}=-3$	A1
	Tangents meet on the axis of symmetry of curve C so $x=\frac{1+4}{2}=\frac{5}{2}$	M1 A1
	When $x=\frac{5}{2}$ $y=-3\left(\frac{5}{2}-1\right)=-\frac{9}{2}$	M1
	$\left(\frac{5}{2},-\frac{9}{2}\right)$	A1 (6)
	Notes	
(b)		
M1	For $\frac{dy}{dx}=2x-5$ (Allow if seen in part(c))	
A1	$\frac{dy}{dx}=-3$ when $x=1$	
M1	For use of $\frac{x_1+x_2}{2}$ or $\frac{-b}{2a}$ or $\frac{dy}{dx}=0$	
A1	$x=\frac{5}{2}$	
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m(x-x_1)$ oe	
A1	For $\left(\frac{5}{2},-\frac{9}{2}\right)$	

Type MS · Finding the equation of a normal

Answer 5 of 14 · Question 6 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 6 · find part c in this mark scheme extract

	Alternative Method $\frac{dy}{dx} = 2x - 5$ When $x = 1$ $\frac{dy}{dx} = -3$ So $y = -3x + 3$ When $x = 4$ $\frac{dy}{dx} = 3$ So $y = 3x - 12$ Meet when $-3x + 3 = 3x - 12$ $\left(\frac{5}{2}, -\frac{9}{2}\right)$	M1 A1 M1 A1 M1 A1 (6)
	Notes	
M1	**Alternative** For an attempt to find the equation of the tangent of C at A	
A1	For $y = -3x + 3$	
M1	For an attempt to find the equation of the tangent of C at B	
A1	For $y = 3x - 12$	
M1	For equating the two equations	
A1	For $\left(\frac{5}{2}, -\frac{9}{2}\right)$	

Type MS · Finding the equation of a normal

Answer 5 of 14 · Question 6 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 6 · find part c in this mark scheme extract

(c)	Gradient of normal at (0, 1) = $\frac{1}{3}$	M1
	When $x=\frac{5}{2}$ $y=\frac{1}{3}\left(\frac{5}{2}-1\right)=\frac{1}{2}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)
	Alternative Method $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$	M1
	Meet when $\frac{1}{3}x-\frac{1}{3}=-\frac{1}{3}x+\frac{4}{3}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)

	Notes
(c)	
M1	For Gradient of normal at (0, 1) = $\frac{1}{3}$ ft the gradient found in part (b)
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m\left(x-x_1\right)$ oe
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$
	Alternative
M1	For $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$
M1	For equating the two equations
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$

Type MS · Finding the equation of a normal

Answer 5 of 14 · Question 6 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 4 of 6 · find part c in this mark scheme extract

(d)	Area $= \frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 1 Area $= \frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 2 $AN=\sqrt{1.5^2+0.5^2}=\frac{\sqrt{10}}{2}$ $AT=\sqrt{1.5^2+4.5^2}=\frac{3\sqrt{10}}{2}$	M1
	Area $= 2\times\frac{1}{2}\times\frac{\sqrt{10}}{2}\times\frac{3\sqrt{10}}{2}=\frac{30}{4}=\frac{15}{2}$	M1 A1 (3)
	Alternative Method 3 Area $= \frac{1}{2}\begin{vmatrix} 1 & \frac{5}{2} & 4 & \frac{5}{2} & 1 \\ 0 & -\frac{9}{2} & 0 & \frac{1}{2} & 0 \end{vmatrix} \Rightarrow \frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
		[15]

	Notes
(d)	
M1	For $\frac{1}{2}\times AB\times NT$
M1	For $\frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)$
A1	For $\frac{15}{2}$
	Alternative 1
M1	For area of triangle ANB + area of triangle ATB
M1	For $\frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}$
A1	$\frac{15}{2}$
	Alternative 2
M1	For finding AN and AT
M1	For $2\times\frac{1}{2}\times AN\times AT$
A1	$\frac{15}{2}$

Type MS · Finding the equation of a normal

Answer 5 of 14 · Question 6 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 5 of 6 · find part c in this mark scheme extract

	Alternative 3
M1	Use of area $= \frac{1}{2}\begin{vmatrix} 1 & a & 4 & c & 1 \\ 0 & b & 0 & d & 0 \end{vmatrix}$ oe where (a, b) and (c, d) are the coordinates of T and N
M1	$\frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)$
A1	$\frac{15}{2}$

Type MS · Finding the equation of a normal

Answer 5 of 14 · Question 6 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 6 of 6 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 6 of 14 · Question 6 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

Number		
6 (a)	$\frac{dy}{dx} = \frac{2}{\sqrt{x}}$	M1
	When $x = 9$ $\frac{dy}{dx} = \frac{2}{3}$	A1
	$y - 12 = \frac{2}{3}(x - 9)$	M1
	When $y = 0$ $-12 = \frac{2}{3}(x - 9)$	M1
	$x = -9$ So $(-9, 0)$	A1 (5)
(b)	Gradient of Normal: $-\frac{3}{2}$	M1
	$y - 12 = -\frac{3}{2}(x - 9)$	M1
	When $y = 0$ $-12 = -\frac{3}{2}(x - 9)$	M1
	$x = 17$ So $(17, 0)$	A1 (4)
(c)	$\frac{1}{2} \times 12 \times 26 = 156$	M1 A1 (2)
		Total 11 marks

Type MS · Finding the equation of a normal

Answer 6 of 14 · Question 6 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt to differentiate y wrt x. Accept $\frac{dy}{dx} = 2x^{-\frac{1}{2}}$ Allow as a minimum $\frac{dy}{dx} = kx^{-\frac{1}{2}}$ where k is a constant $\neq 0$
	A1	For $\frac{dy}{dx} = \frac{2}{3}$
	M1	For a complete method to find the equation of a straight line. If they use $y = mx + c$ then they must reach a value for c for this mark. Accept the equation of the line with their value of $\frac{dy}{dx}$ using the given coordinates.
	M1	For substitution of $y = 0$ to find a value for x
	A1	For $(-9, 0)$
(b)	M1	For gradient of Normal: $-\frac{3}{2}$ Which is the negative reciprocal of their gradient obtained in (a)
	M1	For a complete method to find the equation of a straight line with a gradient of $-\frac{3}{2}$ (ft their gradient of normal) If they use $y = mx + c$ then they must reach a value for c for this mark.
	M1	For substitution of $y = 0$ to find a value for x
	A1	For $(17, 0)$
(c)	M1	For any correct method for finding the area of a triangle. e.g., $A = \frac{1}{2} \times 12 \times ('17' - '-9') = 156$ OR $A = \frac{1}{2}\begin{bmatrix} 9 & -9 & 17 & 9 \\ 12 & 0 & 0 & 12 \end{bmatrix} = \frac{1}{2}[(0+0+204)-(0+0-108)] = 156$
	A1	For 156

Type MS · Finding the equation of a normal

Answer 6 of 14 · Question 6 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 7 of 14 · Question 11 part b · 4 marks

2023 JAN · Paper 02 · find part b in this mark scheme extract

Number		
11 a	When $x=0 \quad y=3$	B1
	When $y=0 \quad 0=4-e^{2x} \Rightarrow \ln 4 = 2x \Rightarrow x = \ln 2$	M1 A1 (3)
b	$\frac{dy}{dx} = -2e^{2x}$	M1
	When $x=\ln 2 \quad \frac{dy}{dx} = -8$	M1
	$y-0=\frac{1}{8}(x-\ln 2) \Rightarrow y=\frac{1}{8}x-\frac{1}{8}\ln 2$	M1 A1 (4)
c	$\int_0^{\ln 2}\left(4-e^{2x}\right)dx=\left[4x-\frac{1}{2}e^{2x}\right]_0^{\ln 2}=\left(4\ln 2-\frac{1}{2}e^{2\ln 2}\right)-\left(0-\frac{1}{2}e^{0}\right)$	M1M1M1
	$\frac{1}{2}(\ln 2)\left(\frac{1}{8}\ln 2\right)=\frac{1}{16}(\ln 2)^2$	M1M1
	$\frac{1}{8}\int_0^{\ln 2}\lvert x-\ln 2\rvert\,dx=\frac{1}{8}\left\lvert\frac{x^2}{2}-x\ln 2\right\rvert_0^{\ln 2}=\left\lvert\frac{(\ln 2)^2}{16}-\frac{(\ln 2)^2}{8}\right\rvert=\frac{(\ln 2)^2}{16}$	[M1M1]
	$4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2=1.3$	M1 A1 (7)
		Total 14 marks

Type MS · Finding the equation of a normal

Answer 8 of 14 · Question 7 part a · 1 mark

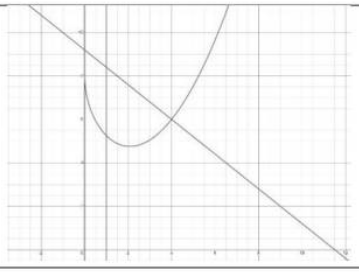
2024 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

7(a)	$(a=)\frac{4^2}{4}-3\sqrt{4}+8=6$	B1 cso [1]
(b)	$\left(\frac{dy}{dx}=\right)\frac{2x}{4}-3\times\frac{1}{2}\times x^{-\frac{1}{2}}$ oe $\left(x=4\Rightarrow\frac{dy}{dx}=\right)\frac{2\times4}{4}-3\times\frac{1}{2}\times4^{-\frac{1}{2}}=\frac{5}{4}\Rightarrow$ $M_n=-\frac{1}{"\frac{5}{4}"}$ $y-6="-\frac{4}{5}"(x-4)\Rightarrow 5y+4x-46=0$ *	M1 dM1A1 M1 (now matches ePen) M1A1 cso [6]
(c)	Area under C $\left(A_C=\int_1^4\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\,dx\right)$ $=\left[\frac{x^3}{3\times4}-\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}+8x\right]_1^4$ $=\left[\frac{4^3}{3\times4}-\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}+8\times4\right]-\left[\frac{1^3}{3\times4}-\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}+8\times1\right]$ $=\frac{61}{4}$ Area under the line $5\times0+4x-46=0\Rightarrow x(=11.5)$ $A=\frac{1}{2}\times("11.5"-4)\times6=\frac{45}{2}$ **ALT** $\int_4^{11.5}\left(-\frac{4}{5}x+\frac{46}{5}\right)dx=\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x\right]_4^{11.5}$ $=\left(-\frac{4}{5\times2}\times11.5^2+\frac{46}{5}\times11.5\right)-\left(-\frac{4}{5\times2}\times4^2+\frac{46}{5}\times4\right)$ Required area $=\left(\frac{61}{4}+\frac{45}{2}=\right)\frac{151}{4}(=37.75)$ oe	M1 M1 A1 M1A1 A1 [6]

Type MS · Finding the equation of a normal

Answer 8 of 14 · Question 7 part a · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

ALT	Line - curve $\int_1^4\left(\left(-\frac{4}{5}x+\frac{46}{5}\right)-\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\right)\mathrm{d}x$ or $\int_1^4\left(-\frac{x^2}{4}+3\sqrt{x}-\frac{4}{5}x+\frac{6}{5}\right)\mathrm{d}x$	
	$\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x-\frac{x^3}{4\times3}+\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}-8x\right]_1^4$ or $\left[-\frac{x^3}{4\times3}+\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}-\frac{4}{5\times2}x^2+\frac{6}{5}x\right]_1^4$	M1
	$\left(-\frac{4}{5\times2}\times4^2+\frac{46}{5}\times4-\frac{4^3}{4\times3}+\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}-8\times4\right)-$	M1
	$\left(-\frac{4}{5\times2}\times1^2+\frac{46}{5}\times1-\frac{1^3}{4\times3}+\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}-8\times1\right)$ or	
	$\left(-\frac{4^3}{4\times3}+\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}-\frac{4}{5\times2}\times4^2+\frac{6}{5}\times4\right)-\left(-\frac{1^3}{4\times3}+\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}-\frac{1}{5\times2}\times4^2+\frac{6}{5}\times1\right)$	
	$=\frac{127}{20}$	A1
	Area under the line	
	$5\times0+4x-46=0\Rightarrow x\left(=11.5\right)\ 5\times y+4\times1-46=0\Rightarrow x\left(=\frac{42}{5}\right)$	
	$A=\frac{1}{2}\times\left("11.5"-1\right)\times"\frac{42}{5}"=\frac{45}{2}\quad A=\frac{1}{2}\times\left("11.5"-1\right)\times"\frac{42}{5}"=\frac{441}{10}$	M1
	ALT $\int_1^{"11.5"}\left(-\frac{4}{5}x+\frac{46}{5}\right)\mathrm{d}x=\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x\right]_1^{"11.5"}$	A1
	$=\left(-\frac{4}{5\times2}\times"11.5"^2+\frac{46}{5}\times"11.5"\right)-\left(-\frac{4}{5\times2}\times1^2+\frac{46}{5}\times1\right)=\frac{441}{10}$	
	$\left(\frac{441}{10}-\frac{127}{20}=\right)\frac{151}{4}$	A1
	Useful diagram 	
		Total 13 marks

Type MS · Finding the equation of a normal

Answer 8 of 14 · Question 7 part a · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	Correct substitution, no errors and shows $a = 6$
(b)	M1	For an attempt to differentiate the given function. Simplification not required. See General Guidance for the definition of an attempt. No power of x to increase.
	dM1	For substituting the value of $x = 4$ into their derivative. Can be implied by sight of $\frac{5}{4}$ Dependent upon the previous method mark.
	A1	For the correct gradient of $\frac{5}{4}$
	M1	For finding the negative reciprocal of their gradient.
	M1	For forming the equation of the normal, using the given value for a and their **changed** gradient (ie not the gradient of the tangent). This is not a dependent mark, but the candidate may not use the gradient of the tangent and the gradient used must come from some differentiation work. If the candidate uses $y = mx + c$, a fully correct rearrangement to find c must be shown (this can be implied by a correct c for their equation) and concluded with the equation of the line written.
	A1 cso	For the correct equation of the line, minimum steps shown, no errors or omissions.
(c)	M1	For an attempt to integrate the given function. See General Guidance for the definition of an attempt. In this question at least one term correctly integrated and no power of t to decrease. Terms do not need to be simplified. Limits don't need to be present or correct.
	M1	For substituting the correct coordinates into their **changed** expression, the correct way around and subtracting. At least one correct substitution in one term of each limit fully shown. Their limits must be correct.
	A1	For the correct area under the curve. A correct answer here will imply M1 M1 A1 if the integration step has been shown and first M1 awarded. Solutions where the integration step has not been shown will score M0 M0 A0.
	M1	For a correct method to find the area of the triangle using their value for the intersection of line $L$with the x-axis . or Fully correct integration for line and substitution of limits of their 11.5 (following any attempt to find where L crosses the x-axis) and 4. May be implied by correct area. Terms do not need to be simplified.
	A1	For the correct area of the triangle
	A1	For the correct area of $\frac{151}{4}$

Type MS · Finding the equation of a normal

Answer 8 of 14 · Question 7 part a · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part a in this mark scheme extract

ALT	M1	For an attempt to integrate either of the functions as given – there **must** be a subtraction sign if listed separately. Also award this mark for integrating $\int_1^4\left(-\frac{x^2}{4}+3\sqrt{x}-\frac{4}{5}x+E\right)\mathrm{d}x \quad E\neq 0$ (Combines the constant terms incorrectly) See General Guidance for the definition of an attempt. In this question at least one term correctly integrated and no power of t to decrease. Terms do not need to be simplified. Limits don't need to be present or correct.
	M1	For substituting the correct coordinates into their **changed** expression, the correct way around and subtracting. At least one correct substitution in one term of each limit fully shown. Their limits must be correct.
	A1	For $\frac{127}{20}$
	M1	For a correct method to find the area of the large triangle using their value for the intersection of line $L$with the x-axis . or Fully correct integration for line and substitution of limits of their 11.5 (following any attempt to find where L crosses the x-axis) and 4. May be implied by correct area. Terms do not need to be simplified.
	A1	For $\frac{441}{10}$
	A1	For the correct area of $\frac{151}{4}$

Where students have combined expression incorrectly as follows:

$\int_1^4\left(\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\pm\left(-\frac{4}{5}x+\frac{46}{5}\right)\right)\mathrm{d}x$ If the constant term is unsimplified, the highest mark will be up to M1 M1 A1 M0 A0 A0 if the work is seen for the first 3 marks as described above.

If they have simplified the constant term to give eg

$\pm\left(\frac{x^2}{4}-3\sqrt{x}+\frac{86}{5}\right)\pm\frac{4}{5}x$ or $\pm\left(\frac{x^2}{4}-3\sqrt{x}+\frac{86}{5}\right)\pm\frac{86}{5}x$

The highest mark will be M1 M0 A0 M0 A0 A0 if the first two terms in the curve and any constant term are integrated as per conditions above.

Type MS · Finding the equation of a normal

Answer 8 of 14 · Question 7 part a · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 9 of 14 · Question 7 part b · 6 marks

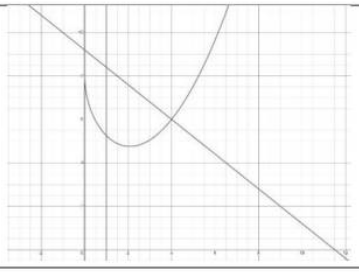
2024 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

7(a)	$(a=)\frac{4^2}{4}-3\sqrt{4}+8=6$	B1 cso [1]
(b)	$\left(\frac{dy}{dx}=\right)\frac{2x}{4}-3\times\frac{1}{2}\times x^{-\frac{1}{2}}$ oe	M1
	$\left(x=4\Rightarrow\frac{dy}{dx}=\right)\frac{2\times4}{4}-3\times\frac{1}{2}\times4^{-\frac{1}{2}}=\frac{5}{4}\Rightarrow$	dM1A1
	$M_n=-\frac{1}{"\frac{5}{4}"}$	M1 (now matches ePen)
	$y-6="-\frac{4}{5}"(x-4)\Rightarrow 5y+4x-46=0$ *	M1A1 cso [6]
(c)	Area under C	
	$\left(A_C=\int_1^4\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\,dx\right)$	
	$=\left[\frac{x^3}{3\times4}-\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}+8x\right]_1^4$	M1
	$=\left[\frac{4^3}{3\times4}-\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}+8\times4\right]-\left[\frac{1^3}{3\times4}-\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}+8\times1\right]$	M1
	$=\frac{61}{4}$	A1
	Area under the line	
	$5\times0+4x-46=0\Rightarrow x(=11.5)$ $A=\frac{1}{2}\times("11.5"-4)\times6=\frac{45}{2}$	
	ALT $\int_4^{11.5}\left(-\frac{4}{5}x+\frac{46}{5}\right)dx=\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x\right]_4^{11.5}$	M1A1
	$=\left(-\frac{4}{5\times2}\times11.5^2+\frac{46}{5}\times11.5\right)-\left(-\frac{4}{5\times2}\times4^2+\frac{46}{5}\times4\right)$	
	Required area $=\left(\frac{61}{4}+\frac{45}{2}=\right)\frac{151}{4}(=37.75)$ oe	A1 [6]

Type MS · Finding the equation of a normal

Answer 9 of 14 · Question 7 part b · 6 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

ALT	Line - curve $\int_1^4\left(\left(-\frac{4}{5}x+\frac{46}{5}\right)-\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\right)\mathrm{d}x$ or $\int_1^4\left(-\frac{x^2}{4}+3\sqrt{x}-\frac{4}{5}x+\frac{6}{5}\right)\mathrm{d}x$	
	$\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x-\frac{x^3}{4\times3}+\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}-8x\right]_1^4$ or $\left[-\frac{x^3}{4\times3}+\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}-\frac{4}{5\times2}x^2+\frac{6}{5}x\right]_1^4$	M1
	$\left(-\frac{4}{5\times2}\times4^2+\frac{46}{5}\times4-\frac{4^3}{4\times3}+\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}-8\times4\right)-$	M1
	$\left(-\frac{4}{5\times2}\times1^2+\frac{46}{5}\times1-\frac{1^3}{4\times3}+\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}-8\times1\right)$ or	
	$\left(-\frac{4^3}{4\times3}+\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}-\frac{4}{5\times2}\times4^2+\frac{6}{5}\times4\right)-\left(-\frac{1^3}{4\times3}+\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}-\frac{1}{5\times2}\times4^2+\frac{6}{5}\times1\right)$	
	$=\frac{127}{20}$	A1
	Area under the line	
	$5\times0+4x-46=0\Rightarrow x\left(=11.5\right)\ 5\times y+4\times1-46=0\Rightarrow x\left(=\frac{42}{5}\right)$	
	$A=\frac{1}{2}\times\left("11.5"-1\right)\times"\frac{42}{5}"=\frac{45}{2}\quad A=\frac{1}{2}\times\left("11.5"-1\right)\times"\frac{42}{5}"=\frac{441}{10}$	M1
	ALT $\int_1^{"11.5"}\left(-\frac{4}{5}x+\frac{46}{5}\right)\mathrm{d}x=\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x\right]_1^{"11.5"}$	A1
	$=\left(-\frac{4}{5\times2}\times"11.5"^2+\frac{46}{5}\times"11.5"\right)-\left(-\frac{4}{5\times2}\times1^2+\frac{46}{5}\times1\right)=\frac{441}{10}$	
	$\left(\frac{441}{10}-\frac{127}{20}=\right)\frac{151}{4}$	A1
	Useful diagram 	
		Total 13 marks

Type MS · Finding the equation of a normal

Answer 9 of 14 · Question 7 part b · 6 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	Correct substitution, no errors and shows $a = 6$
(b)	M1	For an attempt to differentiate the given function. Simplification not required. See General Guidance for the definition of an attempt. No power of x to increase.
	dM1	For substituting the value of $x = 4$ into their derivative. Can be implied by sight of $\frac{5}{4}$ Dependent upon the previous method mark.
	A1	For the correct gradient of $\frac{5}{4}$
	M1	For finding the negative reciprocal of their gradient.
	M1	For forming the equation of the normal, using the given value for a and their **changed** gradient (ie not the gradient of the tangent). This is not a dependent mark, but the candidate may not use the gradient of the tangent and the gradient used must come from some differentiation work. If the candidate uses $y = mx + c$, a fully correct rearrangement to find c must be shown (this can be implied by a correct c for their equation) and concluded with the equation of the line written.
	A1 cso	For the correct equation of the line, minimum steps shown, no errors or omissions.
(c)	M1	For an attempt to integrate the given function. See General Guidance for the definition of an attempt. In this question at least one term correctly integrated and no power of t to decrease. Terms do not need to be simplified. Limits don't need to be present or correct.
	M1	For substituting the correct coordinates into their **changed** expression, the correct way around and subtracting. At least one correct substitution in one term of each limit fully shown. Their limits must be correct.
	A1	For the correct area under the curve. A correct answer here will imply M1 M1 A1 if the integration step has been shown and first M1 awarded. Solutions where the integration step has not been shown will score M0 M0 A0.
	M1	For a correct method to find the area of the triangle using their value for the intersection of line $L$with the x-axis . or Fully correct integration for line and substitution of limits of their 11.5 (following any attempt to find where L crosses the x-axis) and 4. May be implied by correct area. Terms do not need to be simplified.
	A1	For the correct area of the triangle
	A1	For the correct area of $\frac{151}{4}$

Type MS · Finding the equation of a normal

Answer 9 of 14 · Question 7 part b · 6 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

ALT	M1	For an attempt to integrate either of the functions as given – there **must** be a subtraction sign if listed separately. Also award this mark for integrating $\int_1^4\left(-\frac{x^2}{4}+3\sqrt{x}-\frac{4}{5}x+E\right)\mathrm{d}x \quad E\neq 0$ (Combines the constant terms incorrectly) See General Guidance for the definition of an attempt. In this question at least one term correctly integrated and no power of t to decrease. Terms do not need to be simplified. Limits don't need to be present or correct.
	M1	For substituting the correct coordinates into their **changed** expression, the correct way around and subtracting. At least one correct substitution in one term of each limit fully shown. Their limits must be correct.
	A1	For $\frac{127}{20}$
	M1	For a correct method to find the area of the large triangle using their value for the intersection of line $L$with the x-axis . or Fully correct integration for line and substitution of limits of their 11.5 (following any attempt to find where L crosses the x-axis) and 4. May be implied by correct area. Terms do not need to be simplified.
	A1	For $\frac{441}{10}$
	A1	For the correct area of $\frac{151}{4}$

Where students have combined expression incorrectly as follows:

$\int_1^4\left(\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\pm\left(-\frac{4}{5}x+\frac{46}{5}\right)\right)\mathrm{d}x$ If the constant term is unsimplified, the highest mark will be up to M1 M1 A1 M0 A0 A0 if the work is seen for the first 3 marks as described above.

If they have simplified the constant term to give eg

$\pm\left(\frac{x^2}{4}-3\sqrt{x}+\frac{86}{5}\right)\pm\frac{4}{5}x$ or $\pm\left(\frac{x^2}{4}-3\sqrt{x}+\frac{86}{5}\right)\pm\frac{86}{5}x$

The highest mark will be M1 M0 A0 M0 A0 A0 if the first two terms in the curve and any constant term are integrated as per conditions above.

Type MS · Finding the equation of a normal

Answer 9 of 14 · Question 7 part b · 6 marks

2024 JUN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 10 of 14 · Question 7 part b · 7 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

7(a)	$x = -\frac{5}{4}$	B1 [1]
(b)	$\{\frac{dy}{dx}\} = \frac{2x(4x+5) - 4(x^2-1)}{(4x+5)^2}$ $y = 0$ $\{\frac{dy}{dx}\} = \frac{2(-1)(4(-1)+5) - 4((-1)^2-1)}{(4(-1)+5)^2} = -2$ $\rightarrow \{gradient\ of\ normal =\} \frac{1}{2}$ $y - 0 = \frac{1}{2}(x-(-1))oe$	M1A1A1 B1 M1 M1A1cso [7]
(c)	$\frac{x+1}{2} = \frac{x^2-1}{4x+5}$ $2x^2 + 9x + 7 = 0$ ***OR*** $(x+1)(4x+5) = 2(x+1)(x-1)$ $(2x+7)(x+1) = 0 \Rightarrow x = \cdots$ ***OR*** $(4x+5) = 2(x+1) \Rightarrow x = \cdots$ $x = -\frac{7}{2}, \{x = -1\}$ $y = \frac{"-\frac{7}{2}"+1}{2} = \cdots$ $\left(-\frac{7}{2}, -\frac{5}{4}\right)$	M1 dM1 ddM1 A1 M1 A1 [6]
		Total 14 marks

Type MS · Finding the equation of a normal

Answer 10 of 14 · Question 7 part b · 7 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	$x = -\frac{5}{4}$ *or* $x = -1.25$, do not accept 4*x*+5=0
(b)	M1	Attempts to differentiate *y*. **Quotient Rule:** Look for $\frac{Px(4x+5) - Q(x^2-1)}{(4x+5)^2}$, *where* $P > 0, Q > 0$ If the quotient rule is quoted, it must be correct. Condone invisible bracket for this mark. **Product Rule:** Look for $Px(4x+5)^{-1} \pm Q(x^2-1)(4x+5)^{-2}$, *where* $P > 0, Q > 0$
	A1	Quotient rule: Correct denominator $(4x+5)^2$and one correct term in the numerator, 2*x*(4*x*+5) or $-4(x^2-1)$. Product rule: One term correct
	A1	Fully correct. $\left\{\frac{dy}{dx} =\right\} \frac{2x(4x+5)-4(x^2-1)}{(4x+5)^2}$ or $\left\{\frac{dy}{dx} =\right\} 2x(4x+5)^{-1} - 4(x^2-1)(4x+5)^{-2}$oe
	B1	*y* = 0 seen or implied
	M1	Correct method finding gradient of normal to the curve where *x*= -1
	M1	Correct method of forming the equation of a straight line using the point (**-1,their y value**) with a "**changed**" gradient.("changed" means a different gradient from their tangent gradient, their normal could be found by an incorrect method) If using *y*=*mx*+*c*, a value for *c* must be found and the equation formed.
	A1cso	For the correct line in any form, simplified or unsimplified. (isw) Once correct equation is seen, award the mark, ignore later incorrect simplification. e.g.2*y* = *x*+1 or 2*y*-*x*-1=0 or *y*=0.5x+0.5 oe or even $y - 0 = \frac{1}{2}(x - (-1))$
(c)	M1	For equating the equation of their line with the equation of the curve.
	dM1	For forming an equation with all like terms collected. **OR** cross multiplies then factorises e.g. $2(x^2 - 1) = 2(x+1)(x-1)$
	ddM1	Solves their equation with a correct method and finds at least one real value of *x* which is not -1, method of solving could be implied by correct answers if calculator used. **OR** cancels (*x*+1) and solves their linear equation.
	A1	$x = -\frac{7}{2}$, $\{x = -1\}$
	M1	For substituting their *x* value (not -1) to either the equation of their normal or the equation of the given curve to find a value for *y*
	A1	For the correct exact coordinates of $D\left(-\frac{7}{2}, -\frac{5}{4}\right)$ *oe* Allow *x* = -3.5, y = -1.25 instead of given as coordinates.

Type MS · Finding the equation of a normal

Answer 10 of 14 · Question 7 part b · 7 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 11 of 14 · Question 4 part c · 6 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

4 (a)	$f(x)=2(x-3)^2-5$ $\Rightarrow f(x)=2(x^2-6x+9)-5=2x^2-12x+13$ $p=-12,*\ \ q=13$ **ALT** $\frac{dy}{dx}=4x+p=0$ at $(3,-5)$ $\Rightarrow 4\times 3+p=0\Rightarrow p=-12*$ $-5=2\times 3^2-12\times 3+q\Rightarrow q=13$	M1M1 M1A1 cso [4] M1 M1 M1A1 cso [4]
(b)	Minimum since coefficient of x^2 is positive	B1 [1]
(c)	$\frac{dy}{dx}=4x-12\Rightarrow$ when $x=1$ $\frac{dy}{dx}=-8$ So gradient of normal $=\frac{1}{8}$ When $x=1$ $y=3$ $y-'3'=\frac{1}{'8'}(x-1)$ Eg. $y-3=\frac{1}{8}(x-1)$, $x-8y+23=0$ oe	B1 B1ft M1 M1 A1, A1 [6]
		Total 11 marks

Type MS · Finding the equation of a normal

Answer 11 of 14 · Question 4 part c · 6 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using the coordinate of the stationary point and forming the expression: $f(x)=2(x-3)^2-5$ This must be correct for this mark
	M1	Expands their $2(x-3)^2-5$ to form a 3TQ
	M1	For equating coefficients with the given equation to find the value of p and q
	A1	For $p=-12,*\ q=13$ **NB:** The value of p is given in the question.
	ALT – Uses differentiation	
	M1	For a correct differentiation of the expression for y This must be correct for this mark
	M1	For equating their $\frac{dy}{dx}=0$, substituting in $x = 3$ and attempting to find a value for p
	M1	For substituting the given value of p into $y=2x^2+px+q$ using the coordinates $(3,-5)$ to find the value of q. Explicit substitution of the value of p into the equation must be seen if p has not been found using differentiation. [M0M0M1A0 can be scored]
	A1	For $p=-12,*\ q=13$ **NB:** The value of p is given in the question.
(b)	B1	For the correct statement: Minimum since coefficient of x^2 is positive
	ALT	
	B1	For $\frac{d^2y}{dx^2}=4\Rightarrow$ positive, hence minimum. A correct conclusion is required.
(c)	B1	For the value of the gradient when $x = 1$ $[m=-8]$
	B1ft	For the gradient of the normal which must have come from differentiation. Ft their -8
	M1	For the value of y when $x = 1$ using their value of q
	M1	For a complete equation of the line in any form using their normal and their value of y Allow this mark for an equation of the line with $x = 1$ and $y =$ '3' with their normal, provided there is evidence that the normal comes from the negative reciprocal of their value of the tangent. For this mark the gradient of the tangent need not have come from differentiation.
	A1	For the correct equation in any form.
	A1	For the correct equation in the required form or any multiples of the coefficients.

Type MS · Finding the equation of a normal

Answer 11 of 14 · Question 4 part c · 6 marks

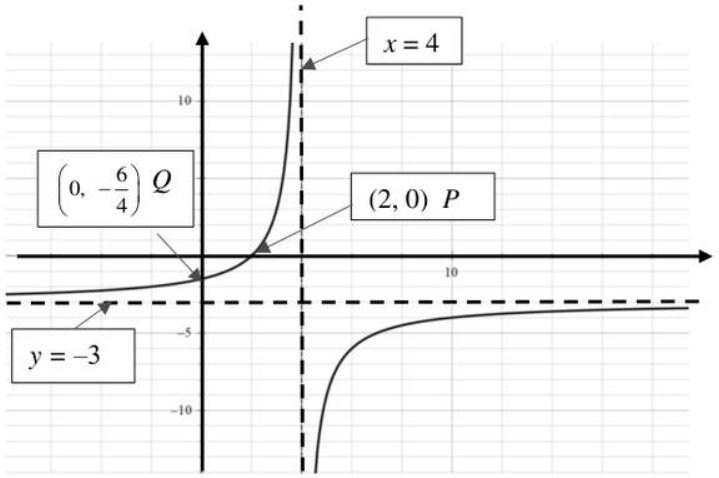
2021 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 12 of 14 · Question 10 part d · 6 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks
10(a) **(i)**	When $y = 0$, $x = 2$ or $(2, 0)$	B1
(ii)	When $x = 0$, $y = -\frac{6}{4}$ oe or $\left(0, -\frac{6}{4}\right)$ oe	B1 [2]
	If not labelled part (i) and (ii), do not award marks unless the candidate has presented in the correct order or has made it clear which coordinate or pair of values is *P* and which is *Q*	
(b)	(i) $x = 4$ (ii) $y = -3$	B1 B1
	If not labelled part (i) and (ii), do not award marks unless the candidate has presented in the correct order or has identification of horizontal (or parallel to *x*-axis) and vertical (or parallel to *y*-axis) is clear.	[2]
(c)		B1 B1 ft B1 ft [3]
(d)	$\left(\frac{dy}{dx} =\right) \frac{-3(x-4)-(6-3x)\times 1}{(x-4)^2} = \left[\frac{6}{(x-4)^2}\right]$ oe	M1A1A1
	where $x = 2$ $\frac{dy}{dx} = \frac{6}{(2-4)^2} = \frac{6}{4} \Rightarrow$ Gradient of normal $= -\frac{4}{6}$ oe	M1
	Equation of normal: $y - 0 = "-\frac{4}{6}"(x-2) \Rightarrow [3y = -2x + 4]$ oe	dM1A1 [6]
(e)	$\frac{6-3x}{x-4} = \frac{-2x+4}{3} \Rightarrow 2x^2 - 21x + 34 = 0$	M1
	$\Rightarrow (2x-17)(x-2) = 0$	M1
	At R $\Rightarrow x = \frac{17}{2}$, (2)	A1 [3]
		Total 16 marks

Type MS · Finding the equation of a normal

Answer 12 of 14 · Question 10 part d · 6 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

Part	Mark	Notes
(a) (i)	B1	Must state $y = 0$, $x = 2$ or $(2, 0)$ or clearly stating $x = 2$
(ii)	B1	Must state $x = 0$, $y = -\frac{6}{4}$ oe or $\left(0, -\frac{6}{4}\right)$ oe or clearly stating $y = -\frac{6}{4}$
(b)(i)	B1	For $x = 4$
(ii)	B1	For $y = -3$
(c)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn and **must not** obviously 'bend back' on themselves. Mark intention.
	B1ft	For the asymptotes drawn, follow through their (b)(i) and (ii). There **must** be at least one branch of a negative reciprocal curve present in the correct place for their work in (a) and (b), which **must** not cross or obviously bend back from the asymptotes. The asymptotes **must** be labelled with their equation or shown as passing through 4 on the x-axis and -3 on the y-axis, both clearly labelled.
	B1ft	For at least one branch of a negative reciprocal curve in the correct place for their work in (a) and (b), passing through their intercepts on the axes. The intercepts should be correctly labelled with the coordinates or the axes labelled correctly with $-\frac{6}{4}$ (oe) and 2 (or their ft values) but condone labelling to be P and Q.
(d)	M1	For an expression of the form. $\frac{-a(x-4)-(6-3x)\times b}{(x-4)^2}$ oe
	A1	For an expression of the form. $\frac{-3(x-4)-(6-3x)\times b}{(x-4)^2}$ or $\frac{-a(x-4)-(6-3x)\times 1}{(x-4)^2}$ .oe
	A1	Fully correct – need not be simplified.
	M1	For substituting $x = 2$ into their $\frac{dy}{dx}$ and finding the gradient of the normal. This is not a dependent method mark, but the substitution must be into a **changed** function. If their expression does not allow substitution of $x = 2$, this mark cannot be awarded.
	dM1	For a complete and correct method to find the equation of the normal using their (changed gradient), $y = 0$ and $x = 2$. Dependent on the previous mark. If $y = mx + c$ is used they must find a value for c.
	A1	For any correct equation. **This can be in any form and may be left unsimplified.**
(e)	M1	For equating their equation of the normal to C and attempting to form a 3TQ. The attempt must involve correctly removing both denominators of the equation as a minimum and any attempt to collect terms.
	M1	For a minimally acceptable (see general guidance) and complete attempt to solve their quadratic equation, leading to a value of x..
	A1	For the x coordinate of point R $x = \frac{17}{2}$

Type MS · Finding the equation of a normal

Answer 12 of 14 · Question 10 part d · 6 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 13 of 14 · Question 8 part whole · 8 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

8	When $(x=\pi \quad y=)2\pi^2$	B1
	$\frac{dy}{dx}=4x-\cos x$	M1
	When $x=\pi \quad \frac{dy}{dx}=4\pi-\cos\pi \quad =4\pi+1$	M1 A1
	Gradient of the normal $=-\frac{1}{"4\pi+1"}$	M1
	$y-2\pi^2=-\frac{1}{4\pi+1}(x-\pi)$ oe	M1
	$\left[(4\pi+1)(y-2\pi^2)=-x+\pi\right]$	
	$4\pi y+y-8\pi^3-2\pi^2-\pi+x=0$	dM1
	$x+(4\pi+1)y-\pi(8\pi^2+2\pi+1)=0 \quad *$	A1 cso
		Total 8 marks

Type MS · Finding the equation of a normal

Answer 13 of 14 · Question 8 part whole · 8 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a normal

Answer 14 of 14 · Question 9 part b · 7 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

9 (a)	$f'(x) = 4 \times 2x(1+x^2)^3 = 8x(1+x^2)^3$ *	M1 A1* cso [2]
(b)	Find the gradient of the tangent to C at A:	
	$\frac{dy}{dx} = \frac{3\cos 3x(1+x^2)^4 - 8x(1+x^2)^3 \sin 3x}{(1+x^2)^8}$	M1A1A1
	$= \frac{(1+x^2)^3[3(1+x^2)\cos 3x - 8x\sin 3x]}{(1+x^2)^8}$	
	$= \frac{3\cos 3x + 3x^2\cos 3x - 8x\sin 3x}{(1+x^2)^5}$	
	$\frac{3\cos\left(3\times\frac{2\pi}{3}\right) + 3\left(\frac{2\pi}{3}\right)^2\cos\left(3\times\frac{2\pi}{3}\right) - 8\times\frac{2\pi}{3}\sin\left(3\times\frac{2\pi}{3}\right)}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^5}$	M1
	$= \frac{3\left(1+\left(\frac{2\pi}{3}\right)^2\right)}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^5} = \frac{3}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4}$	A1
	Find the negative reciprocal:	
	$= -\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$	M1A1* cso [7]
ALT (b)	Find the gradient of the tangent to C at A:	
	Using product rule from	M1
	$y = \sin 3x\left(1+x^2\right)^{-4}$	A1A1
	$\frac{dy}{dx} = \sin 3x\left(-8x\left(1+x^2\right)^{-5}\right) + 3\cos 3x\left(1+x^2\right)^{-4}$	
	$\frac{dy}{dx}\left(\frac{2\pi}{3}\right) = \sin\left(3\times\frac{2\pi}{3}\right)\left(-8\times\frac{2\pi}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^{-5}\right) + 3\times\cos\left(3\times\frac{2\pi}{3}\right)\left(1+\left(\frac{2\pi}{3}\right)^2\right)^{-4}$	M1
	$= \frac{3}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4}$ oe	A1
	Find the negative reciprocal:	
	$= -\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$	M1A1* cso [7]
		Total 9 marks

Type MS · Finding the equation of a normal

Answer 14 of 14 · Question 9 part b · 7 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
9 (a)	M1	For showing an intermediate equation in which are shown multiplication by the original index, multiplication by the derivative of the bracketed term and a reduction in the index of the bracketed term. Should be in the form $ax(1+x^2)^3$
	A1*cso	For a fully correct solution, minimum steps shown, no omissions or error. Accept "$\frac{dy}{dx}=$" or "$\frac{d}{dx}=$"
(b)	M1	For a minimally acceptable attempt to differentiate using the quotient rule • $\sin 3x \Rightarrow k\cos 3x \quad k \neq 0,1$ where k is an integer • $\left(1+x^2\right)^4$ as given in (a) • Allow + /- in the numerator • Denominator must be correct
	A1	For both terms in the numerator, in either order
	A1	For the correct expression shown oe **NB: This mark appears as an M mark in ePen**
	M1	For substitution of $x=\frac{2\pi}{3}$ into any changed expression representing the gradient of the tangent. **NB: This mark appears as an A mark in ePen**
	A1	For the correct expression shown. This may be implied by further correct work. **NB: This mark appears as an M mark in ePen**
	M1	For attempting to find the negative reciprocal of their expression.
	A1cso	For the correct expression shown
ALT (b)	M1	For a minimally acceptable attempt to differentiate using the product rule • $\sin 3x \Rightarrow k\cos 3x \quad k \neq 0,1$ where k is an integer • $\left(1+x^2\right)^4$ as given in (a) • Allow + /- between the terms
	A1	For one term correct
	A1	For the correct expression shown oe **NB: This mark appears as an M mark in ePen**
	M1	For substitution of $x=\frac{2\pi}{3}$ into any changed expression representing the gradient of the tangent. **NB: This mark appears as an A mark in ePen**
	A1	For the correct expression shown This may be implied by further correct work **NB: This mark appears as an M mark in ePen**
	M1	For attempting to find the negative reciprocal of their expression.
	A1cso	For the correct expression shown

Type MS · Finding the equation of a normal

Answer 14 of 14 · Question 9 part b · 7 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks